Short- and long-term morbidity and mortality in the population exposed to dioxin after the "Seveso accident".

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The early effects of 2,3,7,8-tetrachlorodibenzo-para-dioxin (TCDD) exposure in the population involved in the Seveso, Italy, incident in 1976, have been examined in numerous studies. Chloracne was the only effect linked with sufficient certainty to dioxin exposure. The possible long-term consequences were investigated with mortality and cancer incidence studies. Mortality and morbidity findings during the 20-year period following the accident showed increased risk from lymphoemopoietic neoplasm, digestive system cancer (rectum in males, and biliary tract among females, in particular) and respiratory system cancer (lung, among males). In the incidence analyses, also thyroid gland and pleura cancer appeared suggestively increased. Soft tissue sarcomas showed an increase in the largest, yet least exposed, exposure sub-cohort. Several hypotheses associating non-cancer effects with dioxin exposure were corroborated by findings in the Seveso population: this was the case with cardiovascular effects (possibly linked to both chemical exposure and stressful disaster experience), endocrine effects (diabetes among females) and reproductive effects: exposure of men to TCDD was linked to a lowered male/female sex ratio in their offspring. The results of many Seveso studies point to possible gender effects, in accordance with animal models. Notwithstanding the acknowledged study limitations (lack of individual exposure markers, short latency, and small population size for certain cancer types), results of previous experimental and epidemiological studies, along with mechanistic knowledge on dioxin toxicity, support the hypotheses that the observed excesses might be associated with dioxin exposure. The mortality and cancer incidence follow-up of the Seveso cohort are continuing.